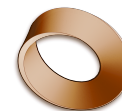


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Mr. Rawson • Y9 MYP Physics



1.2 – Hooke's Law in a Spreadsheet

Unit 1: Oscillations & Waves

Last lesson you drew a line of best fit **by eye**. Today you give a machine the same numbers and see whether it agrees with you. The spreadsheet is not doing the physics — you did that when you measured. It only draws faster, and it never nudges a point to make the line look better.

Learning intentions

- Enter measurements into a spreadsheet with headings and units.
- Make a **scatter** chart from your own data.
- Add a **trendline** and read its equation.
- Decide whether the machine's line agrees with yours — and which you trust.

Part 1 — Set up the sheet

Method

1. Google Drive → **New** → **Google Sheets**. Click *Untitled spreadsheet* at the top left and name it **Hooke's Law — your name**.
2. In cell **A1** type Force (N). In **B1** type Extension (cm).
3. From Part 3 of your Hooke's Law packet, type in the readings for **one** of your three springs — force down column A, extension down column B. **Include your zero row**.
4. **Check before you go on:** numbers sit hard against the *right* of their cell. If any of yours are pushed to the *left*, the sheet has read them as words, not numbers. Retype them.

Why force goes in the first column. Whatever sits in the **leftmost** column becomes the **horizontal** axis — the same way round as the graph you drew by hand, which is the only reason the two can be compared.

Part 2 — The scatter chart

Method

1. Select both columns, headings included: click the **A** at the top, hold **shift**, click the **B**.
2. **Insert** → **Chart**.
3. Sheets will almost certainly pick the *wrong* chart. In the **Chart editor** on the right, under **Setup**, open **Chart type** and choose **Scatter chart** — points only, no joining lines.
4. **Customise** → **Chart & axis titles**: give the chart a title, then label **both** axes with the quantity *and* its unit.

- 1) **State** the title you gave your chart, and your two axis labels.

Model answer: Any sensible title naming both quantities, e.g. Extension of spring 1 against force. Axes must carry a unit: Force / N and Extension / cm. A label with no unit is not a label.

Part 3 — The trendline

Trendline: the spreadsheet's name for a line of best fit.

Method

1. **Chart editor** → **Customise** → **Series**.
2. Tick **Trendline**. Leave the type on **Linear**.
3. Change **Label** to **Use equation**. The equation now appears in the chart's legend.

2) **Copy** the equation of your trendline, exactly as the sheet writes it.

$$y = \text{_____} x + \text{_____}$$

Model answer: Any straight-line equation from real spring data. The x coefficient is the gradient in cm/N — typically between about 2 and 4 cm/N for this spring set. The constant should be small, within a few millimetres of zero.

3) **Compare** this line with the one you drew by hand last lesson. Do they agree?

Model answer: They should lie almost on top of one another. A close match is the expected result and is worth saying plainly.

4) **Explain** which of the two lines you trust more, and why.

Model answer: Either answer earns credit if the reason is sound. The machine is consistent and cannot be tempted to favour a point; but it also fits every point equally, including one you know was a bad reading, which a person would discount. The strongest answers say that the two agreeing is the real reassurance.

5) **Explain** why the number on the end of the equation is not exactly zero, even though a spring with no force on it has no extension.

Model answer: *Measurement uncertainty — reading the natural length, the ruler's position, the spring not hanging quite straight. The line is fitted to real readings that each carry a small error, so it misses the origin slightly. It does not mean the spring was already stretched.*

Name: _____

Date: _____

Part 4 — What the gradient tells you

Gradient: how much the vertical quantity changes for each unit of the horizontal one. Here: the extra centimetres of stretch you get per newton of force.

6) **State** your gradient, with its unit.

gradient = _____ cm/N

Model answer: The x coefficient from Part 3, carried across with the unit cm/N. The unit is the point of the question.

7) **Suggest** what a spring with a *steeper* line would feel like to pull — stiffer, or floppier? Justify your answer using the gradient.

Model answer: Floppier. A steeper line means more centimetres of stretch for each newton, so it gives way more easily. This catches people out: steep looks strong, but on these axes steep means weak.

Extension

Your gradient is in centimetres per newton. The **spring constant** k is that number turned upside down, and in metres:

$$k = \frac{1}{\text{gradient in m/N}} \quad \text{and} \quad \text{m/N} = \text{cm/N} \div 100$$

Your springs are a Pasco *Equal Length Spring Set* — the maker states each one is **25, 30, 35, 40** or **50** N/m, to within $\pm 5\%$. **Which is yours?**

If you finish: add your other two springs as columns D and E, F and G, and put all three on the *same* chart. Which line is steepest? Which spring is stiffest? Make sure your answer to those two is not the same spring.

Word bank: cell column scatter chart trendline gradient equation axis

Keep this sheet. The chart you have just made is the graph that goes into your Hooke's Law write-up next lesson.